THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA FORM THREE PRE-MIDTERM EXAMINATION

BASIC MATHEMATICS

Time: 3 Hours **August 2024**

INSTRUCTIONS

1. This paper consists of sections A and section B with a total of 14 questions.
2. Attempt all questions in both sections.
3. Each question in section A carries 6 marks while each question in section B carries 10 marks.
4. Mathematical tables and non-programmable calculator may be used.
5. Write your examination number on every page of your answer sheets.

SECTION A (60 MARKS)

1. (a) i. Find the average of all factors of 96

   ii. Evaluate: $\frac{2}{5}+\frac{3}{5}\left[\frac{5}{9}+\frac{3}{4}\right]$

   (b) Find the sum of the LCM and GCF of all odd numbers but multiple of 5 between$30-60$

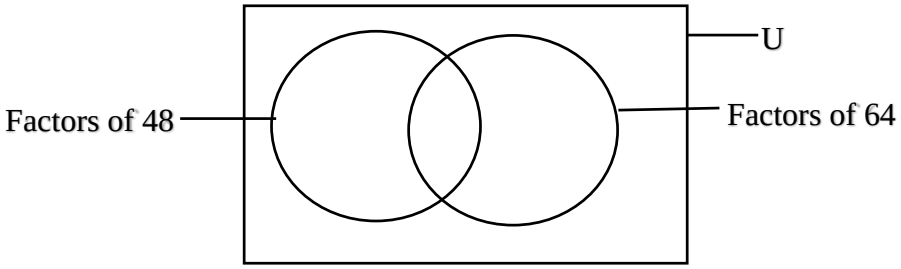
2. (a) i. Solve : $\frac{9^x+9^x+9^x}{3^x+3^x}=13\frac{1}{2}$

   ii. Solve: $b^{-3}=8$

   (b) Solve: $\log_y 200=3\log_3 2+2\log_3 5$

3. (a) In a class of 52 students, 20 study French but not Chinese, 19 study Chinese but not French and 5 study neither of the two languages. Determine the number of students who study either French or Chinese.

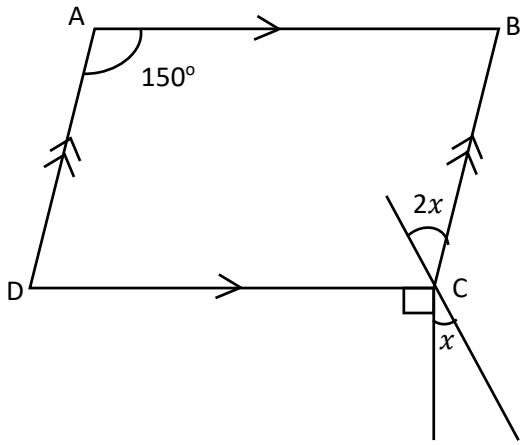
   (b) i. Copy and complete the following Venn diagram to show the factors of 48 and 64



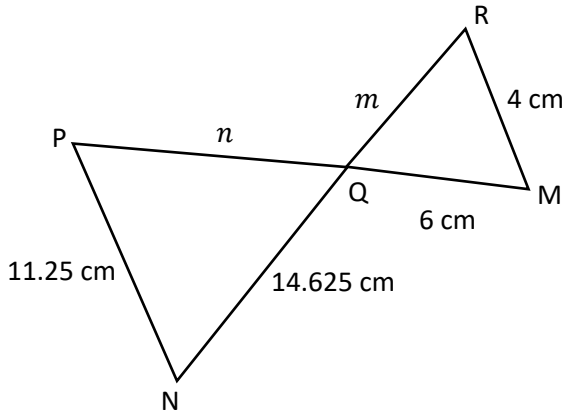
ii. Write the highest common factor of 48 and 64

4. (a) Find the equation of a line whose x −intercept is -5 and y − intercept is 5.

(b) Calculate y in the following figure



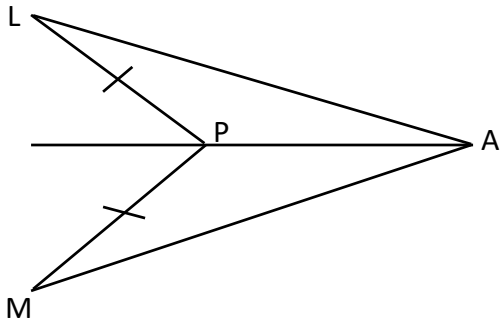
5. (a) The following triangles are similar.



i. Write similar triangles in notation form

ii. Find lengths marked $m \; and \; n$

(b) Use the following figure to prove that $\overline{AP}$ bisects $\angle LAM$, given that$\Delta ALP \cong \Delta AMP$



6. (a) A car travels 60 km using 5 litres of diesel. How many litres of diesel are needed to travel 210 km?

(b) If v varies directly as the square of x and inversely as y and if $v = 18\ when\ x = 3\ and\ y = 4, find\ the\ value\ of\ v\ when x = 5\ and\ y = 2$

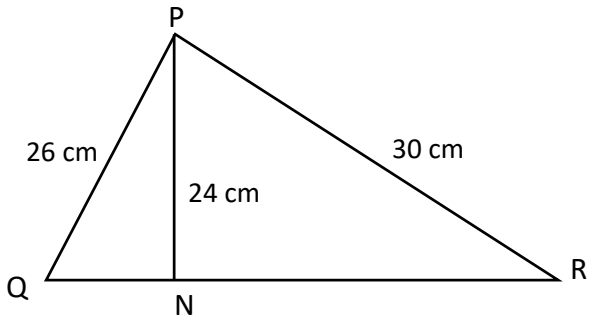
7. (a) Find the percentage profit on an article bought at 20 000 shillings and later sold at 29 000 shillings.

   (b) Find the simple interest on 1 200 000 shillings deposited in a bank which offers an interest rate of $4\frac{1}{2}\%$ per three months if the money stays in the bank for 3 years.

8. (a) Use mathematical tables to calculate the value of $\frac{9\,804 \times 23.19}{0.086 \times 41\,750}$

   (b) When the angle of elevation of the sun is 30° the shadow of the post is 6 m longer than when it is 60°. Find the height of the post

9. (a) In the following figure, prove that $\angle PNR = 90^{\circ}$ and hence calculate $\overline{NR}$



   (b) If $\tan x = \frac{5}{12}$ where x is an acute angle, find $\frac{\sin x + cos x}{2}$

10. (a) i. Use the difference of two square to find the exact value of $8\,999^2 - 7\,001^2$

    ii. Factorize: $x^2 + 7x + 10$

    (b) Solve the following equation by the method of completing the square: $x^2 - 11x + 30 = 0$

SECTION B (40 MARKS)

11. The weight of students at Seypalm secondary school are given as follows:

    71 40 61 43 47 62 63 67 75 51 55 49 41 42 48 65 69 68 66 65 77 63 64 68 68

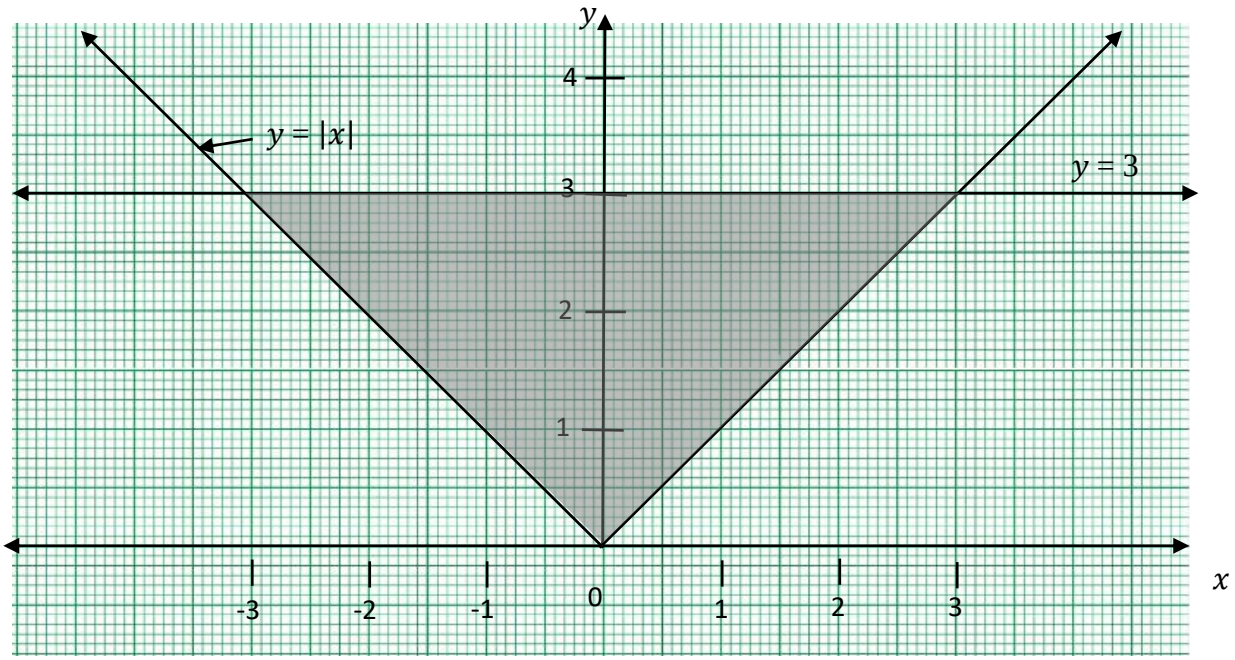
    44 45 56 58 67 67 65 78 59 61 61 62 66 60 60 50 45 63 63 64 64 47 66 66 47

    i. Enter the data into a frequency distribution table using class intervals: $40 - 49$, $50 - 59, and\ so\ on$.

    ii. Calculate the mode

    iii. Calculate the mean by assumed mean of $54\frac{13}{26}$

12. (a) Draw the graph of the inverse of the following graphical relation and hence obtain the domain and range of the inverse of relation



(b) Find the domain and range of the following relation $y = 2x^2 + 1$

13. (a) i. The scale on a road map is 1: 25 000. Two points A and B are 60 cm apart on the map. Find the real distance between them

ii. Find the image of the point $L\ (1, 2)$ under a rotation through 180° clockwise about the origin.

(b) Find the image of point A (3, 2) after a reflection in the line $y = -x$ and sketch both the point and its image

14. (a) Given $f(x) = x^2 - 6x + 9$, find the following:

i. Sketch the graph of f and state the domain and range of f
ii. Line of symmetry
iii. Turning point
iv. Minimum or maximum value of f
v. $x-$ Intercept and $y-$ intercept.

(b) i. Write the step function that demonstrates or describes the following graphical data

ii. State the domain and range of f

